MATH 345 Skeleton Notes
Unit 4: Orthogonality and least squares
Section 4.2: Orthogonal projection

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We introduce orthogonal projection and then use it to derive the Gram–Schmidt algorithm.

Definition and formula

Definition: The orthogonal complement of a subspace

If U is a subspace of $\mathbb{R}^n$, the orthogonal complement of U is the subset of $\mathbb{R}^n$ given by

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{y} = 0 \text{ for all } \mathbf{y} \text{ in } U\}$$

Theorem: Properties of orthogonal complements

The following properties of the orthogonal complement hold:

1. For any subspace U of $\mathbb{R}^n$, $U^\perp$ is a subspace of $\mathbb{R}^n$
2. $\{\mathbf{0}\}^\perp = \mathbb{R}^n$ and $(\mathbb{R}^n)^\perp = \{\mathbf{0}\}$
3. If $U = \text{span}\{\mathbf{x}_1, \dots, \mathbf{x}_k\}$, then

$$U^\perp = \{\mathbf{x} \text{ in } \mathbb{R}^n \mid \mathbf{x} \cdot \mathbf{x}_i = 0 \text{ for } i = 1, 2, \dots, k\}$$

Activity: Finding an orthogonal complement (U4-LO2)

Find $U^{\perp}$ if $U = \operatorname{span}\{(1, 1, 0), (0, 1, 1)\}$ in $\mathbb{R}^3$.

The third property of Properties of orthogonal complements implies that a vector $\mathbf{w} = (x, y, z)$ lies in $U^{\perp}$ if and only if it is orthogonal to both $(1, 1, 0)$ and $(0, 1, 1)$. Taking these two dot products gives

$$\mathbf{w} \cdot (1, 1, 0) = x + y = 0$$
$$\mathbf{w} \cdot (0, 1, 1) = y + z = 0.$$

With columns in the order x, y, z, the augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right].$$

This matrix is already in row-echelon form with leading-one pivots, so Gaussian elimination requires no row operations. The pivot variables are x and y, and z is free. The system is consistent. Back substitution gives $y = -z$ from the

second row, then $x = -y = z$ from the first row. Thus every solution has the form

$$\mathbf{w} = \begin{bmatrix} z \\ -z \\ z \end{bmatrix} = z \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \qquad z \in \mathbb{R},$$

and

$$U^{\perp} = \operatorname{span}\left\{\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}\right\} = \left\{\begin{bmatrix} z \\ -z \\ z \end{bmatrix} : z \in \mathbb{R}\right\}.$$

Fact: Orthogonality after subtracting components

Let $\{\mathbf{q}_1, \mathbf{q}_2, \ldots, \mathbf{q}_m\}$ be an orthonormal set in $\mathbb{R}^n$. Given $\mathbf{x}$ in $\mathbb{R}^n$, define

$$\mathbf{v} = \mathbf{x} - (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 - (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 - \cdots - (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Then $\mathbf{v} \cdot \mathbf{q}_k = 0$ for $k = 1, 2, \ldots, m$. Equivalently, $\mathbf{v}$ lies in the orthogonal complement of the span:

$$\mathbf{v} \in (\operatorname{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_m\})^{\perp}.$$

Why is this true?

Let $c_i = \mathbf{x} \cdot \mathbf{q}_i$ for each i. Given $1 \leq k \leq m$, we calculate that

$$\begin{aligned}\mathbf{v} \cdot \mathbf{q}_k &= (\mathbf{x} - c_1\mathbf{q}_1 - c_2\mathbf{q}_2 - \cdots - c_m\mathbf{q}_m) \cdot \mathbf{q}_k \\ &= \mathbf{x} \cdot \mathbf{q}_k - c_1(\mathbf{q}_1 \cdot \mathbf{q}_k) - \cdots - c_k(\mathbf{q}_k \cdot \mathbf{q}_k) - \cdots - c_m(\mathbf{q}_m \cdot \mathbf{q}_k)\end{aligned}$$

Since the set is orthonormal, $\mathbf{q}_i \cdot \mathbf{q}_k = 0$ for $i \neq k$ and $\mathbf{q}_k \cdot \mathbf{q}_k = 1$. Thus

$$\begin{aligned}\mathbf{v} \cdot \mathbf{q}_k &= \mathbf{x} \cdot \mathbf{q}_k - c_k \\ &= \mathbf{x} \cdot \mathbf{q}_k - \mathbf{x} \cdot \mathbf{q}_k = 0\end{aligned}$$

Remark: When the vector is already in the span

If $\mathbf{x} \in \operatorname{span}\{\mathbf{q}_1, \ldots, \mathbf{q}_m\}$, the expansion theorem (Expansion theorem) gives

$$\mathbf{x} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Thus the vector $\mathbf{v}$ in Orthogonality after subtracting components is $\mathbf{0}$.

Definition: The orthogonal projection onto a subspace

Let U be a subspace of $\mathbb{R}^n$. Then

the orthogonal projection of the vector $\mathbf{x} \in \mathbb{R}^n$ onto U is the vector $\mathbf{y} = \operatorname{proj}_U(\mathbf{x}) \in U$ which is *closest* to $\mathbf{x}$, in the sense that for any $\mathbf{z} \in U$ with $\mathbf{z} \neq \mathbf{y}$, $\|\mathbf{x} - \mathbf{y}\| < \|\mathbf{x} - \mathbf{z}\|$.

The following theorem gives us a practical way to calculate orthogonal projections.

Theorem: Projection theorem

Let U be a subspace of $\mathbb{R}^n$ with orthonormal basis $\{\mathbf{q}_1, \ldots, \mathbf{q}_m\}$. For any vector $\mathbf{x} \in \mathbb{R}^n$, define

$$\mathbf{p} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Then $\mathbf{p} = \text{proj}_U(\mathbf{x})$. The difference

$$\mathbf{v} = \mathbf{x} - \mathbf{p} = \mathbf{x} - \text{proj}_U(\mathbf{x})$$

is called the residual and lies in $U^\perp$. In particular, if $\mathbf{x}$ is already in U, then $\mathbf{x}$ equals $\mathbf{p}$.

Why is this true?

If $\{\mathbf{q}_1, \ldots, \mathbf{q}_m\}$ is an orthonormal basis for U, and we define

$$\mathbf{p} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m,$$

then $\mathbf{p}$ is a linear combination of the basis vectors, so $\mathbf{p} \in U$. By Orthogonality after subtracting components, $\mathbf{x} - \mathbf{p}$ is orthogonal to every basis vector, so $\mathbf{x} - \mathbf{p} \in U^\perp$. Thus if $\mathbf{z} \in U$ and $\mathbf{z} \neq \mathbf{p}$, the Pythagorean theorem (Pythagoras' theorem) tells us that

$$\begin{aligned} \|\mathbf{x} - \mathbf{z}\|^2 &= \|(\mathbf{x} - \mathbf{p}) + (\mathbf{p} - \mathbf{z})\|^2 \\ &= \|\mathbf{x} - \mathbf{p}\|^2 + \|\mathbf{p} - \mathbf{z}\|^2. \end{aligned}$$

Since $\mathbf{p} \neq \mathbf{z}$, we find that

$$\|\mathbf{x} - \mathbf{z}\|^2 > \|\mathbf{x} - \mathbf{p}\|^2,$$

and taking square roots gives that

$$\|\mathbf{x} - \mathbf{z}\| > \|\mathbf{x} - \mathbf{p}\|.$$

Thus the vector $\mathbf{p}$ we have defined is equal to $\text{proj}_U(\mathbf{x})$. Since we have already verified that $\mathbf{x} - \mathbf{p} = \mathbf{x} - \text{proj}_U(\mathbf{x})$ is in $U^\perp$, the theorem follows.

Video walkthrough of Projection theorem: Watch: The projection theorem (1:15).

Orthogonal projection decomposes the original vector into a vector in the subspace and a residual perpendicular to the subspace.

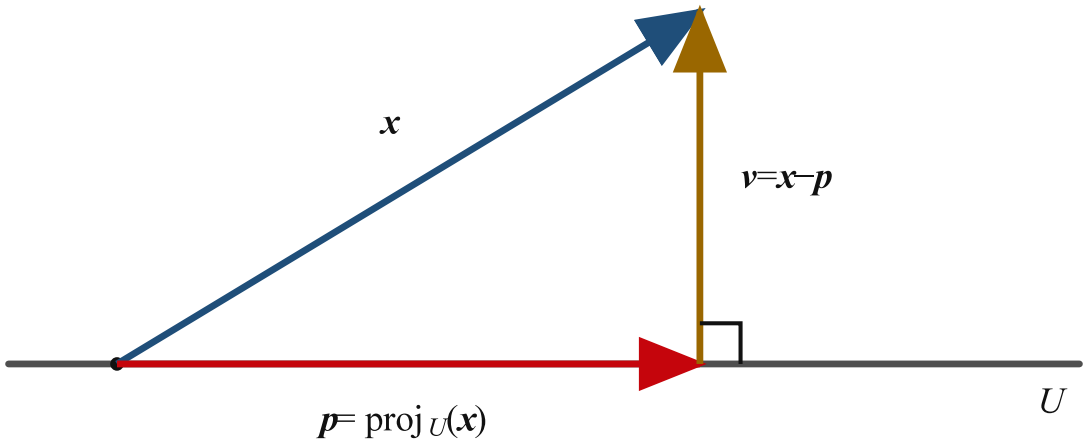


Figure. Projection and residual. Orthogonal projection writes $\mathbf{x} = \mathbf{p} + \mathbf{v}$, where $\mathbf{p} \in U$ and $\mathbf{v} \in U^{\perp}$.

Warning: Normalize before using the projection formula

The formula in Projection theorem requires an *orthonormal* basis: mutual orthogonality alone is not enough. If you are given only an orthogonal set $\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ spanning U, first normalize each vector:

$$\mathbf{q}_i = \frac{\mathbf{f}_i}{\|\mathbf{f}_i\|}, \qquad i = 1, \dots, m.$$

The vectors $\mathbf{q}_i$ form an orthonormal basis of the same subspace. Use these unit vectors in the projection formula.

Note: Expansion versus projection

The expansion theorem and the projection theorem use the same orthonormal basis and the same coefficient formula. The difference in their hypotheses is that the expansion theorem requires $\mathbf{x} \in U = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$, so the sum equals $\mathbf{x}$ itself. The projection theorem allows any $\mathbf{x} \in \mathbb{R}^n$; the sum equals $\operatorname{proj}_U(\mathbf{x})$, which need not

equal $\mathbf{x}$. When $\mathbf{x} \in U$, the projection equals $\mathbf{x}$ and the residual is zero, recovering the expansion theorem.

Example: Projecting onto a span (U4-LO3)

Find $\operatorname{proj}_U \mathbf{x}$ where $U = \operatorname{span}\{(1,1,0),(1,-1,0)\}$ and $\mathbf{x} = (2,0,1)$ in $\mathbb{R}^3$.

The vectors $\mathbf{f}_1 = (1,1,0)$ and $\mathbf{f}_2 = (1,-1,0)$ are already orthogonal, as

$$\mathbf{f}_1 \cdot \mathbf{f}_2 = 1 \cdot 1 + 1 \cdot (-1) + 0 \cdot 0 = 0.$$

Both vectors have length $\sqrt{2}$, so we first normalize them:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad \mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

They form an orthonormal basis for U. The projection coefficients are

$$\mathbf{x} \cdot \mathbf{q}_1 = \frac{2 \cdot 1 + 0 \cdot 1 + 1 \cdot 0}{\sqrt{2}} = \sqrt{2},$$
$$\mathbf{x} \cdot \mathbf{q}_2 = \frac{2 \cdot 1 + 0 \cdot (-1) + 1 \cdot 0}{\sqrt{2}} = \sqrt{2}.$$

Using Projection theorem, we calculate that

$$\begin{aligned}\operatorname{proj}_U \mathbf{x} &= (\mathbf{x}\cdot\mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x}\cdot\mathbf{q}_2)\mathbf{q}_2 \\ &= \sqrt{2}\,\mathbf{q}_1 + \sqrt{2}\,\mathbf{q}_2 \\ &= \begin{bmatrix}1\\1\\0\end{bmatrix} + \begin{bmatrix}1\\-1\\0\end{bmatrix} = \begin{bmatrix}2\\0\\0\end{bmatrix}.\end{aligned}$$

Activity: Projection onto a plane (U4-LO3)

Find $\operatorname{proj}_U \mathbf{x}$ where $U = \operatorname{span}\{(1,0,1),(1,0,-1)\}$ and $\mathbf{x} = (1,2,1)$ in $\mathbb{R}^3$.

The vectors $\mathbf{f}_1 = (1,0,1)$ and $\mathbf{f}_2 = (1,0,-1)$ are orthogonal because

$$\mathbf{f}_1 \cdot \mathbf{f}_2 = 1\cdot 1 + 0\cdot 0 + 1\cdot(-1) = 0.$$

Both have norm $\sqrt{2}$, so normalizing gives an orthonormal basis of U:

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\1\end{bmatrix}, \qquad \mathbf{q}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\0\\-1\end{bmatrix}.$$

The projection coefficients are

$$\mathbf{x}\cdot\mathbf{q}_1 = \frac{1\cdot 1 + 2\cdot 0 + 1\cdot 1}{\sqrt{2}} = \sqrt{2},$$
$$\mathbf{x}\cdot\mathbf{q}_2 = \frac{1\cdot 1 + 2\cdot 0 + 1\cdot(-1)}{\sqrt{2}} = 0.$$

By Projection theorem,

$$\begin{aligned}\mathbf{p} = \operatorname{proj}_U \mathbf{x} &= (\mathbf{x}\cdot\mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x}\cdot\mathbf{q}_2)\mathbf{q}_2\\ &= \sqrt{2}\,\mathbf{q}_1 + 0\,\mathbf{q}_2 = \begin{bmatrix}1\\0\\1\end{bmatrix}.\end{aligned}$$

The projection onto $\operatorname{span}\{\mathbf{q}_2\}$ is the zero vector because $\mathbf{x}\cdot\mathbf{q}_2 = 0$.

Activity: Projection onto a line (U4-LO3)

Let

$$\mathbf{f} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 2 \\ 5 \end{bmatrix}, \qquad L = \operatorname{span}\{\mathbf{f}\}.$$

1. Compute $\operatorname{proj}_L(\mathbf{x})$.
2. Compute the residual $\mathbf{v} = \mathbf{x} - \operatorname{proj}_L(\mathbf{x})$.
3. Check that $\mathbf{v} \cdot \mathbf{f} = 0$.

First, $\|\mathbf{f}\| = \sqrt{3^2 + 4^2} = 5$, so the unit vector

$$\mathbf{q} = \frac{\mathbf{f}}{5} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

forms an orthonormal basis $\{\mathbf{q}\}$ of L. We have $\mathbf{x} \cdot \mathbf{q} = 2(3/5) + 5(4/5) = 26/5$. By Projection theorem,

$$\operatorname{proj}_L(\mathbf{x}) = (\mathbf{x} \cdot \mathbf{q})\mathbf{q} = \frac{26}{25}\begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 78/25 \\ 104/25 \end{bmatrix}.$$

The residual is

$$\mathbf{v} = \begin{bmatrix} 2 \\ 5 \end{bmatrix} - \begin{bmatrix} 78/25 \\ 104/25 \end{bmatrix} = \begin{bmatrix} -28/25 \\ 21/25 \end{bmatrix}.$$

Finally,

$$\mathbf{v} \cdot \mathbf{f} = \left(-\frac{28}{25}\right) 3 + \left(\frac{21}{25}\right) 4 = 0.$$

Activity: Projection onto a line in code (U4-LO3, U4-LO7)

The following code first normalizes the direction vector, then computes the same projection and residual as the previous activity.

```
import numpy as np

f = np.array([3.0, 4.0])
x = np.array([2.0, 5.0])

q = f / np.linalg.norm(f)
p = (x @ q) * q
v = x - p

p, v, v @ f
```

1. Which variable stores the projection?
2. Which variable stores the residual?
3. Why should `v @ f` be close to zero?
4. What does `q = f / np.linalg.norm(f)` accomplish, and why is it needed before computing `(x @ q) * q`?

The variable `p` stores the projection and `v` stores the residual. The value `v @ f` checks whether the residual is orthogonal to the direction of the line, so it should be close to zero. The assignment `q = f / np.linalg.norm(f)` makes `q` a unit vector spanning the same line as `f`. This normalization is needed because the dot-product formula in Projection theorem requires an orthonormal basis. The expression `(x @ q) * q` then computes

$$\operatorname{proj}_L(\mathbf{x}) = (\mathbf{x} \cdot \mathbf{q})\mathbf{q}, \qquad L = \operatorname{span}\{\mathbf{q}\} = \operatorname{span}\{\mathbf{f}\}.$$

Projection as an optimization problem

The closest-point definition of projection naturally leads us to consider projection as an optimization problem.

Activity: Projection as optimization (U4-LO3, U3-LO9)

Let

$$W = \operatorname{span}\left\{\begin{bmatrix}1\\0\end{bmatrix}\right\} \subseteq \mathbb{R}^2, \qquad \mathbf{b} = \begin{bmatrix}2\\3\end{bmatrix}.$$

A general point of W has the form

Activity: Projection as optimization (U4-LO3, U3-LO9) (continued)

$$\mathbf{w}(t) = \begin{bmatrix} t \\ 0 \end{bmatrix}.$$

1. Write the squared distance $g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2$.
2. Compute $g'(t)$ and find the critical point of g.
3. Use $g''(t)$ to classify the critical point. (Recall the second derivative test: if $g'(t_0) = 0$ and g'' is continuous near t_0, then $g''(t_0) > 0$ gives a local minimum, $g''(t_0) < 0$ gives a local maximum, and $g''(t_0) = 0$ is inconclusive; see OpenStax Calculus Volume 1, Section 4.5.)
4. What point of W is closest to $\mathbf{b}$?
5. How does this agree with the figure and Projection theorem?

We have

$$\mathbf{b} - \mathbf{w}(t) = \begin{bmatrix} 2 - t \\ 3 \end{bmatrix},$$

so

$$g(t) = \|\mathbf{b} - \mathbf{w}(t)\|^2 = (2 - t)^2 + 9.$$

Differentiating gives

$$g'(t) = 2(t - 2).$$

The unique critical point is $t = 2$. Since

$$g''(t) = 2 > 0,$$

this critical point is a local minimum. Because g is an upward-opening quadratic, it is also the unique global minimum. Therefore the closest point in W is

$$\widehat{\mathbf{b}} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

This agrees with the figure and Projection theorem: the residual is

$$\mathbf{b} - \widehat{\mathbf{b}} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}.$$

For every $\mathbf{w}(t) \in W$,

$$(\mathbf{b} - \widehat{\mathbf{b}}) \cdot \mathbf{w}(t) = \begin{bmatrix} 0 \\ 3 \end{bmatrix} \cdot \begin{bmatrix} t \\ 0 \end{bmatrix} = 0 \cdot t + 3 \cdot 0 = 0.$$

Therefore the residual is orthogonal to W.

Example: Deriving the projection formula through critical points

Let $\mathbf{x} \in \mathbb{R}^n$, and let $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$ be an orthonormal basis of U. Every vector in U can be written as

$$\mathbf{u}(\mathbf{c}) = \sum_{i=1}^{m} c_i \mathbf{q}_i, \qquad \mathbf{c} = (c_1, \dots, c_m) \in \mathbb{R}^m.$$

Finding the closest vector to $\mathbf{x}$ is therefore equivalent to choosing $\mathbf{c}$ to minimize

$$g(\mathbf{c}) = \|\mathbf{x} - \mathbf{u}(\mathbf{c})\|^2.$$

Squaring the distance does not change which vector is closest, and it gives a polynomial that is easy to differentiate.

Expand the squared norm using the dot product. Since the basis vectors are orthonormal, the cross terms between distinct basis vectors vanish and each $\|\mathbf{q}_i\|^2 = 1$. Thus

$$\begin{aligned} g(\mathbf{c}) &= \left(\mathbf{x} - \sum_{i=1}^{m} c_i \mathbf{q}_i\right) \cdot \left(\mathbf{x} - \sum_{i=1}^{m} c_i \mathbf{q}_i\right) \\ &= \|\mathbf{x}\|^2 - 2\sum_{i=1}^{m} c_i (\mathbf{x} \cdot \mathbf{q}_i) + \sum_{i=1}^{m} c_i^2. \end{aligned}$$

The partial derivatives are

$$\frac{\partial g}{\partial c_i} = 2c_i - 2(\mathbf{x} \cdot \mathbf{q}_i), \qquad i = 1, \dots, m.$$

Setting them all equal to zero gives the unique critical point $\mathbf{c}^*$, whose entries are

$$c_i^* = \mathbf{x} \cdot \mathbf{q}_i, \qquad i = 1, \dots, m.$$

To see that this critical point is the unique global minimum, "complete the squares". Using $c_i^* = \mathbf{x} \cdot \mathbf{q}_i$, we obtain

$$
\begin{aligned}
g(\mathbf{c}) &= \|\mathbf{x}\|^2 + \sum_{i=1}^{m} \left(c_i^2 - 2c_i c_i^*\right) \\
&= \|\mathbf{x}\|^2 + \sum_{i=1}^{m} \left(c_i^2 - 2c_i c_i^* + (c_i^*)^2 - (c_i^*)^2\right) \\
&= \|\mathbf{x}\|^2 - \sum_{i=1}^{m} (c_i^*)^2 + \sum_{i=1}^{m} (c_i - c_i^*)^2 \\
&= g(\mathbf{c}^*) + \sum_{i=1}^{m} (c_i - c_i^*)^2.
\end{aligned}
$$

In the last equality, we used $g(\mathbf{c}^*) = \|\mathbf{x}\|^2 - \sum_{i=1}^{m} (c_i^*)^2$, obtained by substituting $\mathbf{c} = \mathbf{c}^*$ into the expanded formula for g. The added terms are nonnegative and all vanish exactly when $\mathbf{c} = \mathbf{c}^*$. Consequently, the closest vector in U is

$$
\mathbf{p} = \mathbf{u}(\mathbf{c}^*) = \sum_{i=1}^{m} (\mathbf{x} \cdot \mathbf{q}_i)\mathbf{q}_i = \operatorname{proj}_U(\mathbf{x}).
$$

This recovers the formula in Projection theorem by minimizing squared distance.

Directional derivatives as projection coefficients

Let $\mathbf{v}$ be a unit vector. By Projection theorem,

$$
\operatorname{proj}_{\operatorname{span}\{\mathbf{v}\}}\big(\nabla f(\mathbf{a})\big) = \big(\nabla f(\mathbf{a}) \cdot \mathbf{v}\big)\mathbf{v}.
$$

By Directional derivatives from the gradient,

$$
\operatorname{proj}_{\operatorname{span}\{\mathbf{v}\}}\big(\nabla f(\mathbf{a})\big) = D_{\mathbf{v}} f(\mathbf{a})\mathbf{v}.
$$

Thus $D_{\mathbf{v}} f(\mathbf{a})$ is the signed projection coefficient of the gradient in the oriented direction $\mathbf{v}$. Its absolute value is the length of the projection.

Activity: Directional derivative as a projection coefficient (U4-LO3, U3-LO8)

Recall the gradient and unit vectors from Alignment with the gradient (U3-LO8, U1-LO1):

$$\nabla f(\mathbf{a}) = \begin{bmatrix} 3 \\ 4 \end{bmatrix},$$

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix},$$
$$\mathbf{v}_3 = \frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}, \qquad \mathbf{v}_4 = -\frac{1}{5}\begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

For each unit vector $\mathbf{v}_i$, compute

$$\text{proj}_{\text{span}\{\mathbf{v}_i\}}\big(\nabla f(\mathbf{a})\big) = D_{\mathbf{v}_i} f(\mathbf{a})\mathbf{v}_i.$$

1. Which projection coefficients are positive, and which are negative?
2. Why do $\mathbf{v}_3$ and $\mathbf{v}_4$ give the same projection vector but opposite directional derivatives?

From Alignment with the gradient (U3-LO8, U1-LO1),

$$D_{\mathbf{v}_1} f(\mathbf{a}) = 3, \qquad D_{\mathbf{v}_2} f(\mathbf{a}) = 4,$$
$$D_{\mathbf{v}_3} f(\mathbf{a}) = 5, \qquad D_{\mathbf{v}_4} f(\mathbf{a}) = -5.$$

Therefore,

$$\operatorname{proj}_{\operatorname{span}\{\mathbf{v}_1\}}\big(\nabla f(\mathbf{a})\big) = 3\mathbf{v}_1 = \begin{bmatrix} 3 \\ 0 \end{bmatrix},$$

$$\operatorname{proj}_{\operatorname{span}\{\mathbf{v}_2\}}\big(\nabla f(\mathbf{a})\big) = 4\mathbf{v}_2 = \begin{bmatrix} 0 \\ 4 \end{bmatrix},$$

and

$$\operatorname{proj}_{\operatorname{span}\{\mathbf{v}_3\}}\big(\nabla f(\mathbf{a})\big) = 5\mathbf{v}_3 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

Since $\mathbf{v}_4 = -\mathbf{v}_3$,

$$\operatorname{proj}_{\operatorname{span}\{\mathbf{v}_4\}}\big(\nabla f(\mathbf{a})\big) = (-5)\mathbf{v}_4 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}.$$

The vectors $\mathbf{v}_3$ and $\mathbf{v}_4$ span the same line, so projection onto that line gives the same vector. The directional derivatives have opposite signs because the two unit vectors give opposite orientations of the line.

Connection: score, then combine

Projection and attention both use dot products to score directions and

then use the resulting numbers in a linear combination.
Let

$$Q = \begin{bmatrix} \mathbf{q}_1 & \cdots & \mathbf{q}_k \end{bmatrix}$$

have orthonormal columns. Then

$$Q^T\mathbf{x} = \begin{bmatrix} \mathbf{q}_1^T\mathbf{x} \\ \vdots \\ \mathbf{q}_k^T\mathbf{x} \end{bmatrix}$$

collects the directional coefficients of $\mathbf{x}$. Because each $\mathbf{q}_i$ has length 1, these dot products are the coordinates used in the projection. If $\mathbf{x}$ also has length 1, they are cosine similarities with the directions $\mathbf{q}_i$.
Multiplying by Q combines the directions using those coefficients:

$$Q(Q^T\mathbf{x}) = \sum_{i=1}^{k} (\mathbf{q}_i^T\mathbf{x})\mathbf{q}_i = \mathrm{proj}_{\mathrm{col}(Q)}(\mathbf{x}).$$

Thus projection follows the pattern

$$\text{score with } Q^T \quad \longrightarrow \quad \text{combine with } Q.$$

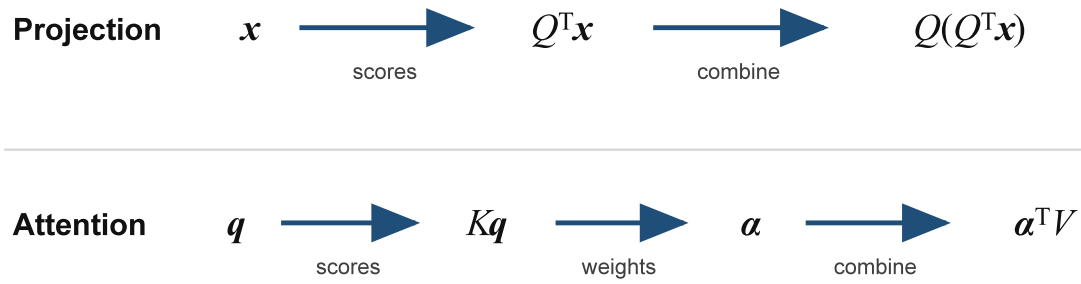


Figure. Score, then combine. Projection uses $Q^T\mathbf{x}$ to score orthonormal directions and $Q(Q^T\mathbf{x})$ to combine them. Attention uses query-key scores, converts them into weights, and combines value vectors.

Attention has the same broad pattern. Query-key dot products produce scores, a weighting rule converts the scores into weights, and the weights

combine value vectors. Projection scores and combines the same orthonormal directions. Attention scores key vectors and then combines value vectors, which may be different vectors. Projection also chooses its coefficients so that the residual is orthogonal to the subspace. Attention is therefore not generally an orthogonal projection, but both computations are built from dot products and linear combinations.

Activity: Dot-product scores as projection coordinates (U4-LO1, U4-LO3)

Let

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\\0\end{bmatrix}, \qquad \mathbf{q}_2 = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\-1\\0\end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix}2\\0\\1\end{bmatrix},$$

and let

$$Q = \begin{bmatrix}\mathbf{q}_1 & \mathbf{q}_2\end{bmatrix}.$$

1. Verify that the columns of Q are orthonormal.
2. Compute the score vector

$$\mathbf{c} = Q^T\mathbf{x}.$$

3. Compute

$$\widehat{\mathbf{x}} = Q\mathbf{c}.$$

4. Compute the residual

$$\mathbf{v} = \mathbf{x} - \widehat{\mathbf{x}},$$

and verify that $Q^T\mathbf{v} = \mathbf{0}$.

Activity: Dot-product scores as projection coordinates (U4-LO1, U4-LO3) (continued)

5. Explain why the entries of $\mathbf{c}$ can be read both as dot-product scores and as coordinates of the projection.
6. Explain how the calculation follows the pattern "score, then combine." What extra condition identifies $\hat{\mathbf{x}}$ as an orthogonal projection?

The two columns have length 1, and

$$\mathbf{q}_1^T \mathbf{q}_2 = \frac{1}{2}(1 - 1) = 0.$$

Thus the columns of Q are orthonormal.
The score vector is

$$\mathbf{c} = Q^T \mathbf{x} = \begin{bmatrix} \mathbf{q}_1^T \mathbf{x} \\ \mathbf{q}_2^T \mathbf{x} \end{bmatrix} = \begin{bmatrix} \sqrt{2} \\ \sqrt{2} \end{bmatrix}.$$

Combining the columns of Q using these scores gives

$$\hat{\mathbf{x}} = Q\mathbf{c} = \sqrt{2}\mathbf{q}_1 + \sqrt{2}\mathbf{q}_2 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}.$$

The residual is

$$\mathbf{v} = \mathbf{x} - \hat{\mathbf{x}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore

$$Q^T\mathbf{v} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Each entry of $\mathbf{c}$ is a dot product with one of the unit directions, so it is both a directional score and a coordinate used to reconstruct the projection. The calculation first scores $\mathbf{x}$ with Q^T, then combines the columns of Q using those scores. The condition $Q^T\mathbf{v} = \mathbf{0}$ says that the residual is orthogonal to every column of Q. This is the extra condition that identifies $\hat{\mathbf{x}}$ as the orthogonal projection.

Gram-Schmidt orthonormalization

The Gram-Schmidt algorithm converts a basis into an orthonormal basis of the same subspace through a sequence of projection, subtraction, measurement, and normalization steps.

Fact: Expanding an orthonormal set

Let $\{\mathbf{q}_1, \dots, \mathbf{q}_m\}$ be an orthonormal set in $\mathbb{R}^n$, and let $\mathbf{x} \in \mathbb{R}^n$ lie outside its span. Define

$$\mathbf{v} = \mathbf{x} - (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 - \cdots - (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m.$$

Then $\mathbf{v} \neq \mathbf{0}$. Normalize it by setting

$$\mathbf{q}_{m+1} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

The enlarged set $\{\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{q}_{m+1}\}$ is an orthonormal set, and the span is preserved:

$$\operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{q}_{m+1}\} = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{x}\}.$$

Why is this true?

If $\mathbf{v} = \mathbf{0}$, then $\mathbf{x} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m$ lies in the original span, contrary to the hypothesis. Thus $\mathbf{v} \neq \mathbf{0}$ and we can normalize it. By Orthogonality after subtracting components, $\mathbf{v}$ is orthogonal to every $\mathbf{q}_k$. The vector $\mathbf{q}_{m+1}$ has norm one and remains orthogonal to every $\mathbf{q}_k$, so the enlarged set is orthonormal.
The formula for $\mathbf{q}_{m+1}$ expresses it as a linear combination of $\mathbf{q}_1, \dots, \mathbf{q}_m, \mathbf{x}$. Conversely,

$$\mathbf{x} = (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + \cdots + (\mathbf{x} \cdot \mathbf{q}_m)\mathbf{q}_m + \|\mathbf{v}\|\mathbf{q}_{m+1}.$$

Each set therefore spans the other, proving the equality of spans.

Activity: Extending an orthonormal set to a basis (U4-LO2)

First normalize the orthogonal vectors

$$\mathbf{f}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{f}_2 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

to obtain an orthonormal set $\{\mathbf{q}_1, \mathbf{q}_2\}$ in $\mathbb{R}^3$. Use Expanding an orthonormal set with $\mathbf{x} = (1, 1, 2)$ to extend this set to an orthonormal basis $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$.

Both given vectors have norm $\sqrt{2}$, so we first set

$$\mathbf{q}_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$$

$$\mathbf{q}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

Their dot product is zero, so they form an orthonormal set. We calculate

$$\mathbf{x} \cdot \mathbf{q}_1 = \frac{1+1}{\sqrt{2}} = \sqrt{2},$$
$$\mathbf{x} \cdot \mathbf{q}_2 = \frac{1-1}{\sqrt{2}} = 0.$$

Using Orthogonality after subtracting components, we obtain

$$\begin{aligned}\mathbf{v} &= \mathbf{x} - (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 - (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 \\ &= \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \sqrt{2}\left(\frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}\right) - 0\mathbf{q}_2 = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}.\end{aligned}$$

Since $\|\mathbf{v}\| = 2$, normalize once more:

$$\mathbf{q}_3 = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

By Orthogonality after subtracting components, $\mathbf{q}_3$ is orthogonal to $\mathbf{q}_1$ and $\mathbf{q}_2$. Since $\|\mathbf{q}_3\| = 1$, the set $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ is orthonormal. These three vectors are linearly independent by Linear in-

dependence of orthogonal sets, so they form an orthonormal basis of $\mathbb{R}^3$.

We use an original column $\mathbf{a}_j$, its projection $\mathbf{p}_j$, the orthogonal residual $\mathbf{v}_j$, the new unit direction $\mathbf{q}_j$, and coefficients r_{ij}. Repeatedly applying Expanding an orthonormal set to the normalized vectors $\mathbf{q}_i$ gives the algorithm below.

Algorithm: Gram-Schmidt algorithm

Input: A matrix

$$A = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{bmatrix}$$

with linearly independent columns.
For $j = 1, \ldots, n$:

1. Current column. Start with $\mathbf{a}_j$.
2. Project. For each $i < j$, compute

$$r_{ij} = \mathbf{q}_i^T \mathbf{a}_j.$$

   Then assemble

$$\mathbf{p}_j = \sum_{i<j} r_{ij} \mathbf{q}_i.$$

   For the first column, $\mathbf{p}_1 = \mathbf{0}$.

3. Subtract.

$$\mathbf{v}_j = \mathbf{a}_j - \mathbf{p}_j.$$

4. Measure.

$$r_{jj} = \|\mathbf{v}_j\|.$$

5. Normalize.

$$\mathbf{q}_j = \frac{\mathbf{v}_j}{r_{jj}}.$$

Output: The vectors $\mathbf{q}_1, \dots, \mathbf{q}_n$ are orthonormal, and

$$\operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_j\} = \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$$

at every stage.

Practice Gram-Schmidt geometrically in Perfectly Normal.

Warning

If $r_{jj} = 0$, then $\mathbf{a}_j$ is already in the span of the earlier columns. The input columns are not linearly independent, and no new unit direction is produced.

The following theorem summarizes the properties of the vectors produced by Gram-Schmidt algorithm.

Theorem: Properties of the Gram-Schmidt output

Apply Gram-Schmidt algorithm to a matrix $A = [\mathbf{a}_1 \ \cdots \ \mathbf{a}_n]$ with linearly independent columns, and let $U = \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$. For each $k = 1, \dots, n$, write $U_k = \operatorname{span}\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$. The output through step k satisfies:

- The span is preserved:

$$\operatorname{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \operatorname{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\} = U_k.$$

- The normalized vectors $\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$ form an orthonormal basis of U_k.

In particular, $\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$ is an orthonormal basis of U.

Example: Gram–Schmidt for a column space (U4-LO2)
Apply the Gram-Schmidt algorithm to the columns of A, where

$$A = \begin{bmatrix} 1 & 3 & 1 \\ 1 & 2 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}.$$

At each stage, record the projection coefficients, projection, residual, residual length, and normalized direction. Give an orthogonal basis and an orthonormal basis for $\operatorname{col}(A)$ (recall Column and Row Space).

1. Compute $\mathbf{v}_1$, r_{11}, and $\mathbf{q}_1$.
2. Compute r_{12}, $\mathbf{p}_2$, $\mathbf{v}_2$, r_{22}, and $\mathbf{q}_2$.
3. Compute r_{13}, r_{23}, $\mathbf{p}_3$, $\mathbf{v}_3$, r_{33}, and $\mathbf{q}_3$.
4. Give an orthogonal basis for $\operatorname{col}(A)$ from the residuals.
5. Give an orthonormal basis for $\operatorname{col}(A)$ from the unit vectors.
6. Explain why both bases span $\operatorname{col}(A)$.

Let the columns of A be

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix},$$

$$\mathbf{a}_2 = \begin{bmatrix} 3 \\ 2 \\ 0 \\ 1 \end{bmatrix},$$

$$\mathbf{a}_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}.$$

For the first column,

$$\mathbf{v}_1 = \mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix},$$

$$r_{11} = \|\mathbf{v}_1\| = 2,$$

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{r_{11}} = \frac{1}{2}\begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}.$$

For the second column,

$$r_{12} = \mathbf{q}_1^T \mathbf{a}_2 = 2,$$

$$\mathbf{p}_2 = r_{12}\mathbf{q}_1 = \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix},$$

$$\mathbf{v}_2 = \mathbf{a}_2 - \mathbf{p}_2 = \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix},$$

$$r_{22} = \|\mathbf{v}_2\| = \sqrt{10},$$

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{r_{22}} = \frac{1}{\sqrt{10}} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}.$$

For the third column,

$$r_{13} = \mathbf{q}_1^T \mathbf{a}_3 = 0,$$
$$r_{23} = \mathbf{q}_2^T \mathbf{a}_3 = \frac{3}{\sqrt{10}},$$
$$\mathbf{p}_3 = r_{13}\mathbf{q}_1 + r_{23}\mathbf{q}_2 = \frac{3}{10}\begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix},$$
$$\mathbf{v}_3 = \mathbf{a}_3 - \mathbf{p}_3 = \frac{1}{10}\begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix},$$
$$r_{33} = \|\mathbf{v}_3\| = \frac{\sqrt{110}}{10},$$
$$\mathbf{q}_3 = \frac{\mathbf{v}_3}{r_{33}} = \frac{1}{\sqrt{110}}\begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix}..$$

An orthogonal basis from the residuals is

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \frac{1}{10} \begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix} \right\}.$$

The corresponding orthonormal basis is

$$\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\} = \left\{ \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ -1 \\ -1 \end{bmatrix}, \frac{1}{\sqrt{10}} \begin{bmatrix} 2 \\ 1 \\ 1 \\ 2 \end{bmatrix}, \frac{1}{\sqrt{110}} \begin{bmatrix} 4 \\ -3 \\ 7 \\ -6 \end{bmatrix} \right\}.$$

The columns $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$ span $\mathrm{col}(A)$. By Properties of the Gram-Schmidt output, Gram-Schmidt preserves each successive span. Therefore the residuals form an orthogonal basis of $\mathrm{col}(A)$, and their normalized unit vectors form an orthonormal basis of $\mathrm{col}(A)$.

Activity: The closest point on a plane (U4-LO3)

Find the point in the plane defined by the equation $x - y - 2z = 0$, which is closest to the point $(1, 2, -1)$.

Let U be the plane. Solving its equation for x gives $x = y + 2z$. If we set $y = s$ and $z = t$, then every vector in U has the form

$$\begin{bmatrix} s + 2t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Therefore

$$U = \operatorname{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Let

$$\mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{u}_2 = \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}.$$

Applying the Gram-Schmidt process from Gram-

Schmidt algorithm gives

$$\mathbf{v}_1 = \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$$

$$\mathbf{v}_2 = \mathbf{u}_2 - \frac{\mathbf{u}_2 \cdot \mathbf{v}_1}{\|\mathbf{v}_1\|^2}\mathbf{v}_1$$

$$= \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} - \frac{2}{2}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

Thus $\{\mathbf{v}_1, \mathbf{v}_2\}$ is an orthogonal basis of U.
First normalize this basis:

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix},$$

$$\mathbf{q}_2 = \frac{\mathbf{v}_2}{\|\mathbf{v}_2\|} = \frac{1}{\sqrt{3}}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}.$$

For the given point $\mathbf{x} = (1, 2, -1)$,

$$\mathbf{x} \cdot \mathbf{q}_1 = \frac{3}{\sqrt{2}}, \qquad \mathbf{x} \cdot \mathbf{q}_2 = -\frac{2}{\sqrt{3}}.$$

Applying Projection theorem to this orthonormal basis gives

$$\begin{aligned} \text{proj}_U(\mathbf{x}) &= (\mathbf{x} \cdot \mathbf{q}_1)\mathbf{q}_1 + (\mathbf{x} \cdot \mathbf{q}_2)\mathbf{q}_2 \\ &= \frac{3}{\sqrt{2}}\mathbf{q}_1 - \frac{2}{\sqrt{3}}\mathbf{q}_2 \\ &= \frac{3}{2}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{2}{3}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5/6 \\ 13/6 \\ -2/3 \end{bmatrix}. \end{aligned}$$

Hence the point in the plane closest to $(1, 2, -1)$ is $(5/6, 13/6, -2/3)$.